Using Fossil Fuels

Chapter 3

Big Question

What are some costs and benefits of using fossil fuels?

People burn fossil fuels to release the energy they contain. We rely on fossil fuels to generate the energy we use in our everyday lives. Look under the hood of an automobile running on gasoline or diesel fuel. You will see and hear the moving parts and feel the heat rising from the engine.

There are many costs and benefits of burning so much fossil fuel on such a large scale. There are also costs and benefits in how these fuels are extracted from Earth's crust. And there are costs and benefits in how the resources are transported, processed, and transformed into usable energy. All this activity accounts for a large part of the world's economy.

It is hard to imagine life without these fuels. But our use of them is relatively recent. On the scale of Earth's history, using fossil fuels has had a major impact on the environment. Fossil fuels are nonrenewable. The costs of using them give us reasons to consider using renewable energy resources.



This petroleum refinery produces many different products. Some will power vehicles. Others will become components of chairs, food containers, drinking straws, roads, and toys.

Extracting Petroleum

Petroleum is found only in some locations on Earth's surface. But most of what has not already been extracted is trapped under layers of rock. Some of this rock is on land, and some is on the ocean floor. After geologists identify an area that might contain petroleum, engineers use drilling technology to break through the rock and get it.

Before drilling through the rock, engineers must understand how much pressure the oil is under. They need to figure out how that pressure will be controlled. The weight of a thousand meters of seawater and many meters of rock on top of all that petroleum can cause oil to explode out of the crust unless the well is controlled. This means that all of the equipment that is going to extract, control, and gather the oil must be in place and ready to work before the extraction begins. These are some of the costs of drilling for oil.



There are many costs to drilling for oil. Dangerous spills can release oil into the ground or large bodies of water.

Transporting Petroleum

Once an oil deposit is tapped, the oil needs to be collected and transported. If the rig is on land or close to shore, a pipeline can carry the oil to a refinery. But a spill from a pipeline can damage the environment. To some people, allowing a pipeline to be built is a chance to make money and add jobs to the local economy. This is a benefit. To others, a pipeline represents an environmental risk that is too great a cost.

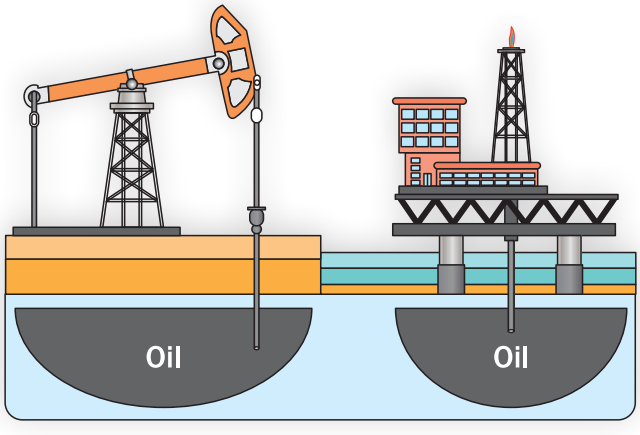
Oil rigs that are far out to sea rely on oil tankers. These are ships that take oil from a rig and carry it elsewhere. But oil tankers can experience accidents, too. Oil tankers have run aground, capsized, sunk, and been captured by pirates. All these can add to the costs of transporting oil.



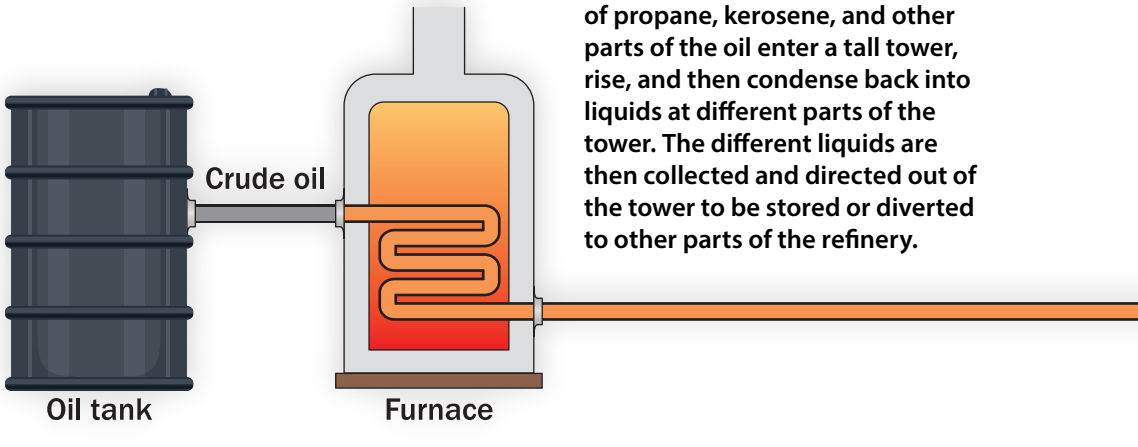
This pipeline carries oil across desert terrain. The benefit is that it moves oil quickly from one place to another. If it fails, one cost is that it can do a lot of damage to the environment.

Refining Petroleum

Petroleum refinement separates petroleum into different forms of liquid oil. The different products can then be burned to generate mechanical, electrical, or thermal energy. Or, they can be turned into petroleum products, such as plastics and asphalt.



The process begins with the removal of oil from the ground. Wells pump oil from beneath land or offshore from beneath the ocean floor.



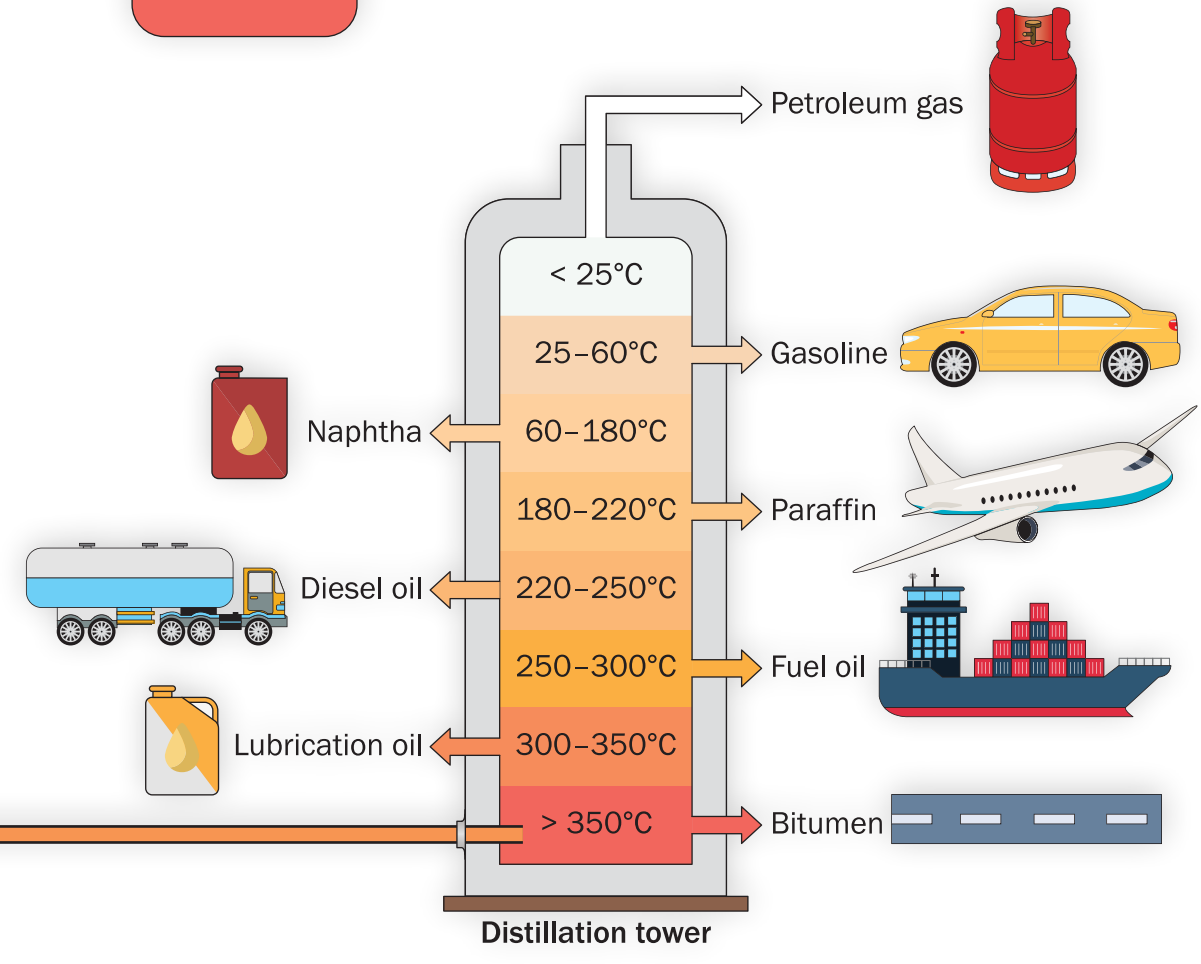
Next, the oil is heated. Different components of the oil boil at different temperatures. Vapors of propane, kerosene, and other parts of the oil enter a tall tower, rise, and then condense back into liquids at different parts of the tower. The different liquids are then collected and directed out of the tower to be stored or diverted to other parts of the refinery.

Gases and **thin oil** collect at the top.

Liquid and **thicker oil** collect in the middle.

The stickiest oil stays at the bottom.

The separated liquids can also be transported to other locations to be processed into other products. For example, some of the components of the synthetic rubber in a basketball are produced by refining petroleum. Nylon, a plastic that is used in clothing and countless other products, is another product of the petrochemical industry.



Coal Mining

Coal is a solid that must be mined. This means it must be dug out of the ground. Coal continues to provide heat for homes and cooking, as well as heating iron ore to produce metal objects. These are some of the benefits of coal.

There are many costs to coal mining as well. It is one of the most dangerous jobs. Mines can be filled with poisonous or explosive gases. They can also collapse, trapping miners. When washed or burned, coal can cause pollution. Coal can be transported by train, truck, or ship. These are less likely to result in an environmental disaster. However, some coal dust can escape into the environment if coal is not properly transported and stored.



The photo shows West Virginia coal miners descending into a mine in 1938. They used shovels to load the car with coal by hand. These days, mining operations use more machinery, which reduces the physical stress on coal miners.

Extracting, Transporting, and Refining Natural Gas

Natural gas is often found where petroleum is found. It is common for the extraction of the two fuels to occur together. A drilling operation may first draw out the gas at the top and then proceed to remove the liquid below. Like petroleum, natural gas can be directed into a pipeline or onto a large ship to transport the gas for processing.

Large pipelines transport large quantities of gas across regions. Smaller networks of pipes carry gas to individual buildings. Many homes or businesses use natural gas for heat. This is one benefit of natural gas. Natural gas also pollutes less than other fossil fuels, but it still pollutes. Typically, gas-powered heating systems have underground gas lines connected to a main line that runs under a street. Burying the pipes helps protect them from being damaged. A broken pipe can become a fiery hazard if the gas is ignited. This can be both dangerous and costly.



Cooling natural gas until it condenses into a liquid allows a large ship to carry about 600 times more methane than it could if the substance were still a gas.

Burning Fossil Fuels

Burning fossils fuels has many costs and benefits. Fossil fuels can be turned into many different products, but mostly they are burned for energy. They help heat our homes and cook our food. They also help power engines and provide electricity.

When fossil fuels are burned, the chemical parts react with oxygen in the air and become new substances. The chemical products of the reaction are mostly released into the atmosphere. This affects Earth's systems.

Burning any fossil fuel produces carbon dioxide. Carbon dioxide is a **greenhouse gas**. Greenhouse gases trap heat on Earth's surface. The more carbon dioxide there is in the atmosphere, the hotter the planet becomes overall. This is a serious cost of burning fossil fuels. Many scientists have stated that large-scale burning of fossil fuels is causing Earth to get warmer. Carbon dioxide that dissolves in water also makes water more acidic, which can make life difficult for animals such as corals and clams.

Vocabulary

greenhouse gas, n. a gas that traps heat in Earth's atmosphere



Greenhouse gases in the atmosphere are warming Earth's overall temperature. The thick ice of Earth's coldest places is melting as the climate warms.